

Lingjun Guo

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Github Page

EDUCATION

Lehigh University

Ph.D. Student. Department of Industrial and Systems Engineering(ISE)
Advisors: Frank Curtis, and Daniel Robinson

Bethlehem, PA

Aug. 2023 – present

University of Michigan, Ann Arbor

M.E., Department of Industrial and Operations Engineering(IOE)

Ann Arbor, MI

Aug. 2021 – Jan. 2023

Fudan University

Bachelor of Science, Department of Mathematics

Shanghai, China

Aug. 2016 – July. 2020

PUBLICATION

- Jianhao Ma, Lingjun Guo, and Salar Fattahi. Behind the scenes of gradient descent: A trajectory analysis via basis function decomposition. In *The Eleventh International Conference on Learning Representations*, 2023. [\[Link\]](#)
- Frank E. Curtis, Lingjun Guo, and Daniel P. Robinson. Progressively Sampled Equality-Constrained Optimization. *Submitted soon.* [\[Link\]](#)

SKILLS

- **Theory:** Neural Network, Large Language Model, Attention Mechanism, Agents, Unconstrained Optimization Algorithms, Complexity Analysis, Probability.
- **Software:** Python(pytorch, torchvision), MATLAB(Deeplearning/Optimization Toolbox).

RESEARCH PROJECTS

Progressive Sampling for Equality-Constrained Optimization

Sep. 2024 - Jun. 2025

ISE, Lehigh University

Advisor: Frank Curtis, Daniel Robinson

- Proposed, analyzed and implemented an algorithm that achieves at least **33% fewer** sample calculations for solving a physics-informed neural network (PINN) problem.
- Analyzed the complexity of the proposed algorithm; showed that it **converges linearly** under mild assumptions.
- Numerically demonstrate the proposed algorithm achieves at least **50% fewer** sample calculations for problems satisfying those assumptions.
- Numerically demonstrate the generalizability of the proposed algorithm, which outperforms competitors in general neural network models that do not fully satisfy those assumptions; achieves $\geq$ **33% fewer** sample calculations.
- **Key words:** constrained optimization, progressively sampling, Sequential Quadratic Programming, PINN.

A Trajectory Analysis Framework for Deep Neural Networks

Jan. 2022 - Oct. 2022

IOE, University of Michigan, Ann Arbor

Advisor: Salar Fattahi, Jianhao Ma

- Proposed a **novel** basis decomposition approach for analyzing learning algorithms for training neural networks. Analyzed learning trajectories when training neural networks. Accepted by ICLR 2023.
- Analyzed learning trajectories for symmetric tensor decomposition via the proposed novel approach. Matched the **best** complexity result in literature.
- Modified network structures of AlexNet, VGG-11, ResNet-18/34/50, and ViT to test proposed approach. Successfully verified our approach on gradient-based learning algorithms(SGD, AdamW, LARS) across popular datasets (CIFAR-10/100), for various choices of batch-size and loss function.
- **Key words:** basis decomposition, over-parametrized model, small initialization, gradient descent, symmetric matrix/tensor decomposition, Deep Neural Networks, layer-wise modification, Pythorch, Colab.

TALKS AND SERVICES

- "Progressively Sampled Equality-Constrained Optimization." Contributed talk in MOPTA 2025, Portugal.
- Reviewer for Mathematical Programming, 2025.